មហាវិទ្យាល័យវិស្វកម្ម
FACULTY OF ENGINEERING

Computer Programming

Lecture 4
Values and Variables - 2

Chhoeum Vantha, Ph.D.
Telecom & Electronic Engineering

Contents

Comments

- Comments are used in C++ to make the code more readable and to explain what the code is doing. The compiler ignores all comments when the program is compiled.

Comments

Types of Comments:

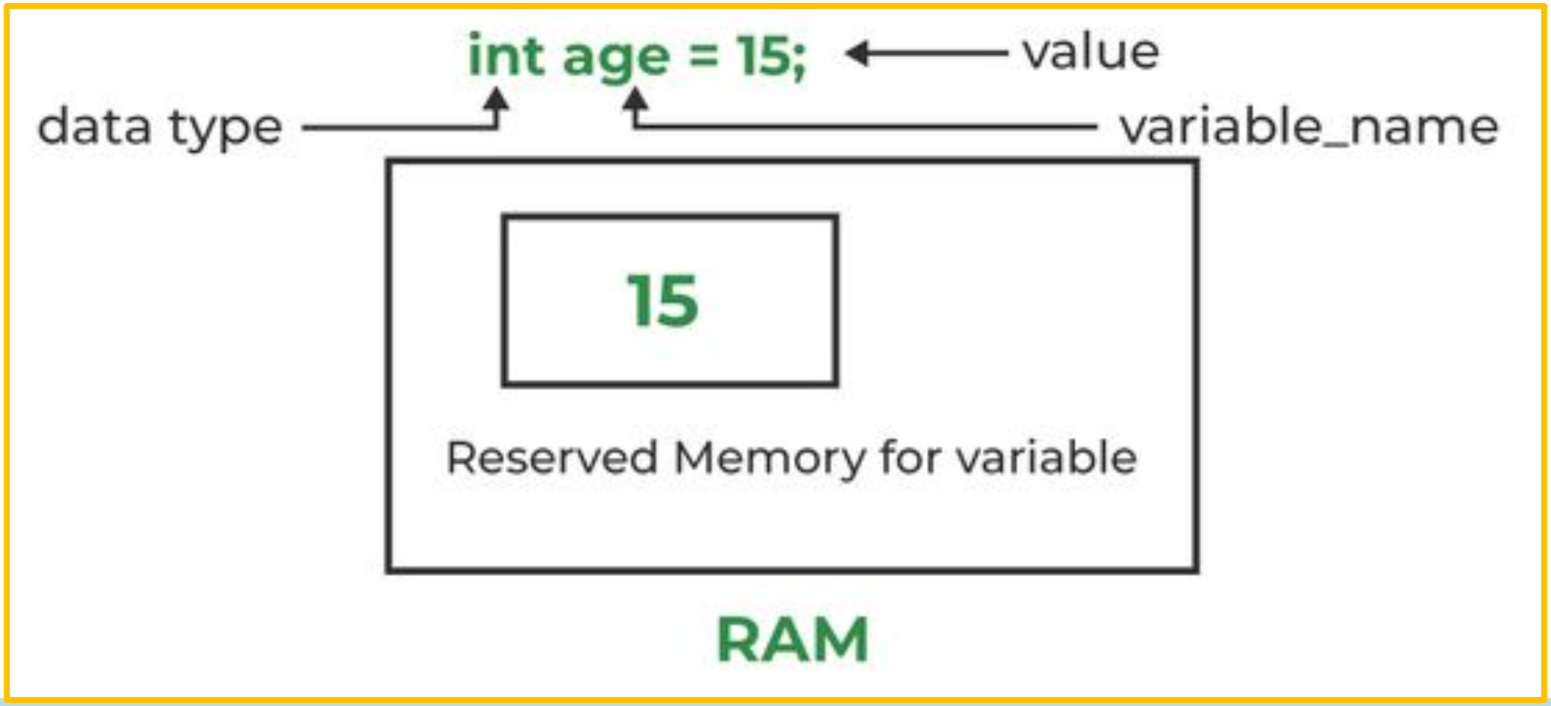
1. Single-line Comment
   - Starts with //
   - Used for short explanations or notes
2. Multi-line Comment
   - Starts with /* and ends with */
   - Used for longer explanations or to temporarily disable code

Comments

```cpp
#include <iostream>
using namespace std;

int main() {
    // Declare variables
    int a = 10;
    int b = 20;

    /* Calculate the sum of a and b
       and print the result */
    int sum = a + b;
    cout << "Sum = " << sum << endl;

    return 0; // Exit the program
}
```

Variable

- A variable is a name given to a memory location. It is used to store data that can be changed during program execution.



Basic Data Types

Data Type	Description	Example Values
int	Integer values	-5, 0, 100
float	Floating-point (decimal) numbers	3.14, -0.1
double	Double-precision float	3.14159, 2.71828
char	Single character	'A', 'z', '3'
bool	Boolean (true/false)	true, false
void	No value (used for functions)	-

Basic Data Types

```cpp
#include <iostream>
using namespace std;

int main() {
    // Integer type
    int age = 20;

    // Floating point type
    float height = 170.5f;

    // Double precision floating point type
    double weight = 65.75;

    // Character type
    char grade = 'A';

    // Boolean type
    bool isPassed = true

```

```cpp
    // Displaying the values
    cout << "Age: " << age << endl;
    cout << "Height: " << height << " cm" << endl;
    cout << "Weight: " << weight << " kg" << endl;
    cout << "Grade: " << grade << endl;
    cout << "Passed Exam: " << isPassed << endl;

    return 0;
}
```

Common Type Modifiers

- Signed, Unsigned, Short, long

Modifier	Description	Example Usage
signed	Default for integer types; can store both positive and negative numbers.	signed int a = -10;
unsigned	Only stores non-negative numbers (0 and positive).	unsigned int b = 20;
short	Reduces the size of integer to at least 16 bits.	short int c = 1000;
long	Increases the size of integer to at least 32 bits.	long int d = 100000L;
long long	Further increases size, usually at least 64 bits.	long long int e = 10000000000LL;
const	Declares a constant value that cannot be changed.	const int f = 50;

check the size of basic data types

- ***sizeof*** operator use to check the size of basic data types in C++

```cpp
#include <iostream>
using namespace std;
int main() {
    cout << "Size of char: " << sizeof(char) << " byte" << endl;
    cout << "Size of int: " << sizeof(int) << " bytes" << endl;
    cout << "Size of short int: " << sizeof(short int) << " bytes" << endl;
    cout << "Size of long int: " << sizeof(long int) << " bytes" << endl;
    cout << "Size of long long int: " << sizeof(long long int) << " bytes" << endl;
    cout << "Size of float: " << sizeof(float) << " bytes" << endl;
    cout << "Size of double: " << sizeof(double) << " bytes" << endl;
    cout << "Size of long double: " << sizeof(long double) << " bytes" << endl;
    cout << "Size of bool: " << sizeof(bool) << " byte" << endl;

    return 0;
}
```

Operations

- Numeric Operators

Operator	*Name*	*Example*	*Result*
+	Addition	34 + 1	35
-	Subtraction	34.0 – 0.1	33.9
*	Multiplication	300 * 30	9000
/	Division	1.0 / 2.0	0.5
%	Modulus	20 % 3	2

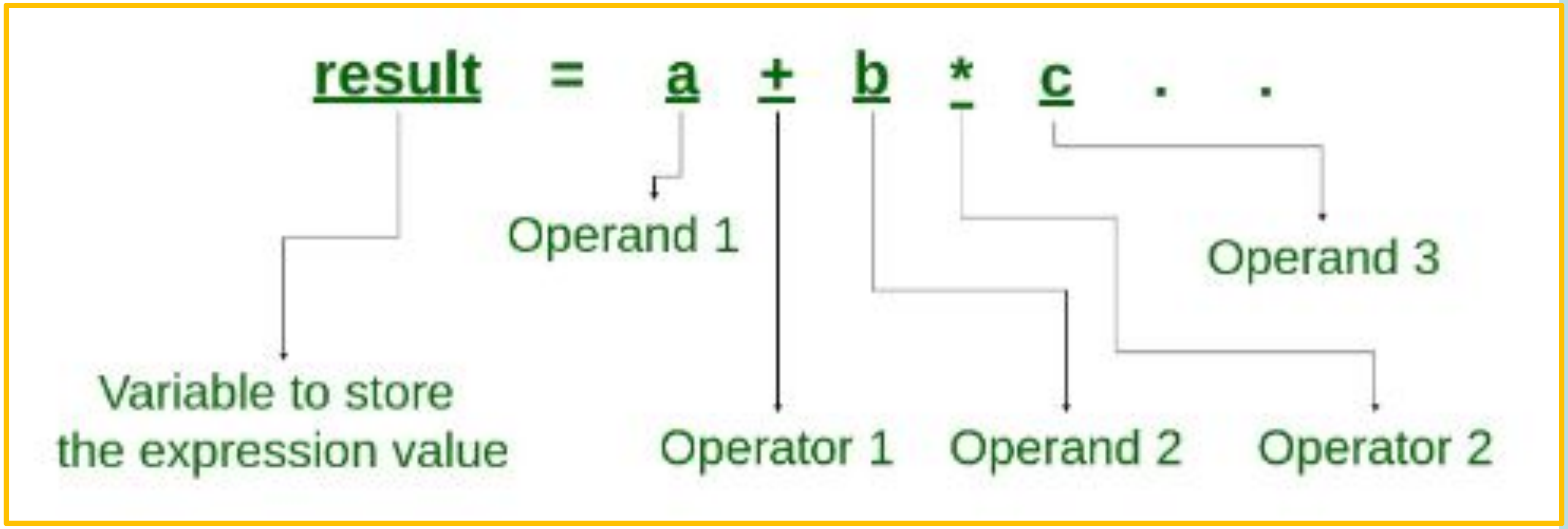
Expressions

Commonly used Python arithmetic binary operators

Expression	Meaning
$x + y$	x added to y, if x and y are numbers x concatenated to y, if x and y are strings
$x - y$	x take away y, if x and y are numbers
$x * y$	x times y, if x and y are numbers x concatenated with itself y times, if x is a string and y is an integer y concatenated with itself x times, if y is a string and x is an integer
x / y	x divided by y, if x and y are numbers

Expressions

- An expression is a combination of operators, constants, and variables.
- An expression may consist of one or more operands, and zero or more operators to produce a value.



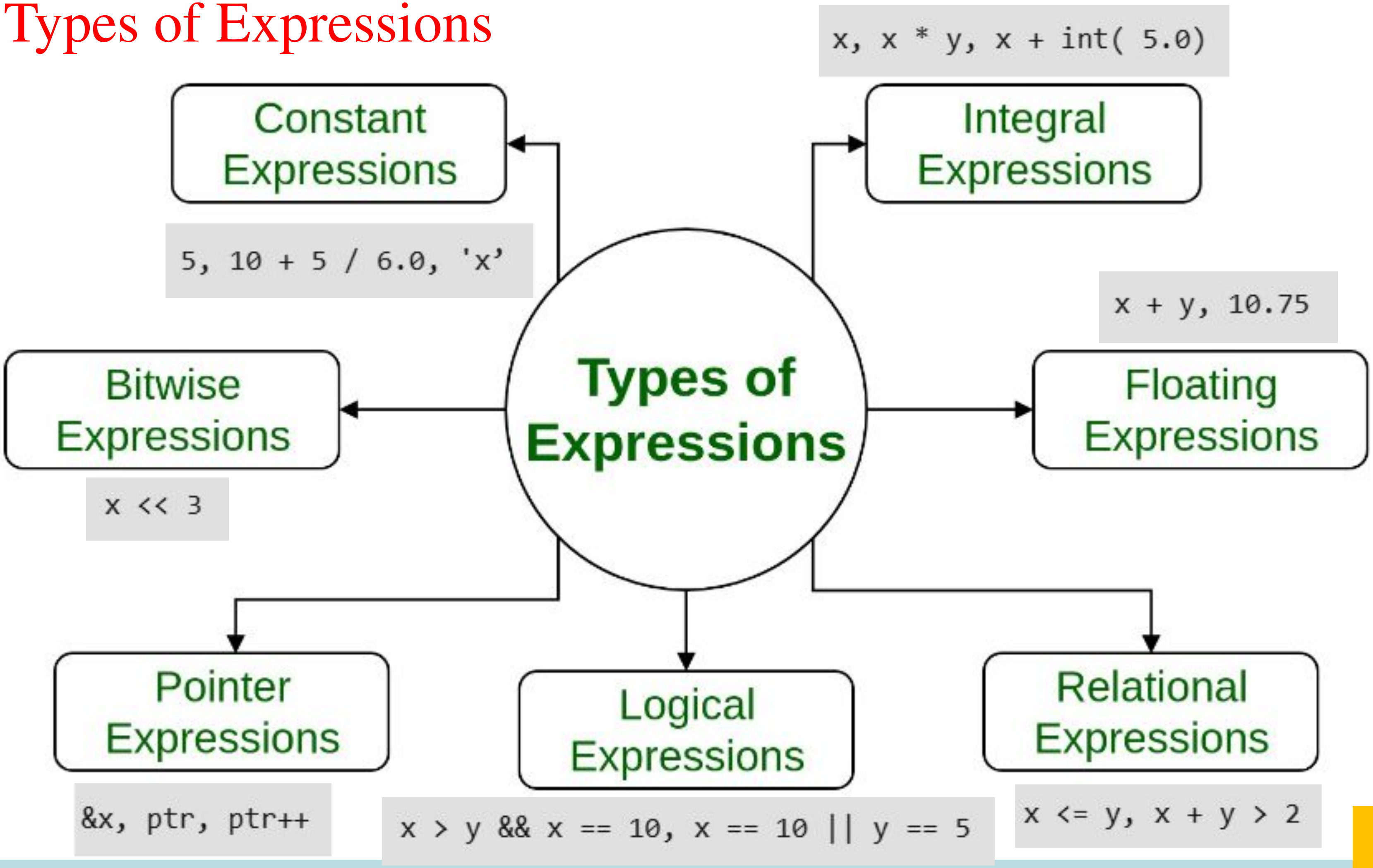
Expressions

```cpp
#include <iostream>
using namespace std;
int main()
{
    int x = 4, y = 8;
    int sum = x + y;
    cout << x << " + " << y << " = " << sum << endl;
    return 0;
}
```

```
4 + 8 = 12
```

Expressions

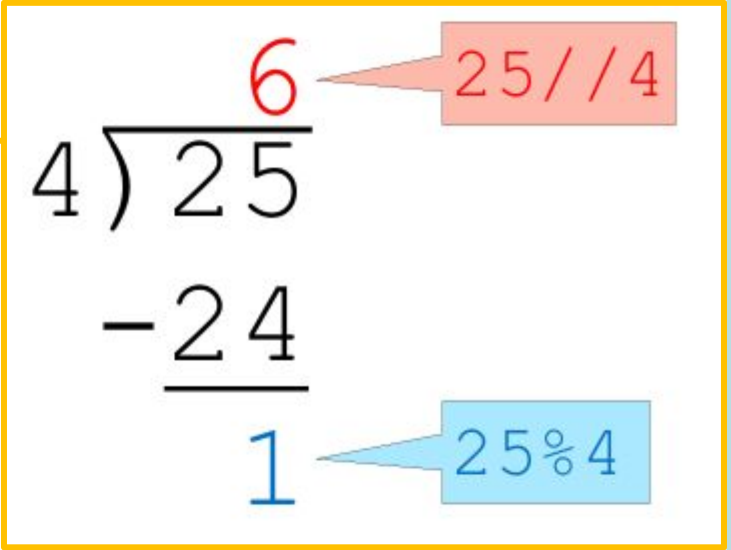
Types of Expressions



Expressions

Integer division and modulus

```
// modulus operator (%)
cout << "25/4 = " << 25/4 <<endl;
cout << "25%4 = " << 25%4 <<endl;
```




```
25/4 = 6
25%4 = 1
```

The modulus operator (%) computes the remainder of integer division

Expressions

The / operator applied to two integers produces a floating-point result

```
//  operator (/)
cout << "25/4 = " << 25/4 <<endl;
cout << "4/25 = " << 4/25 <<endl;
```

```
25/4 = 6
4/25 = 0
```

Mixed Type Expressions

- Expressions may contain mixed integer and floating-point elements
- When an operator has mixed operands—one operand an integer and the other a floating-point number—the interpreter treats the integer operand as a floating-point number and performs floating-point arithmetic

```
// Mixed Type Expressions
int x1 = 4;
float y1 = 10.2;
float s = x1 + y1;
cout << "sum: " << s << endl;
```

```
sum: 14.2
```

W4 - Practice

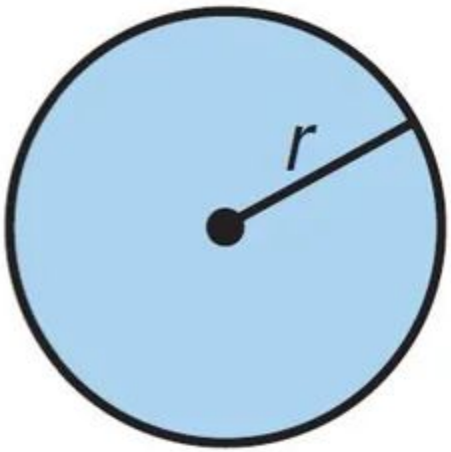
Exercise 1.

Write a program that calculates and displays the area and circumference of a circle with a radius 5. Use float type and pi = 3.14.

radius r

$C = 2\pi r$

$A = \pi r^2$

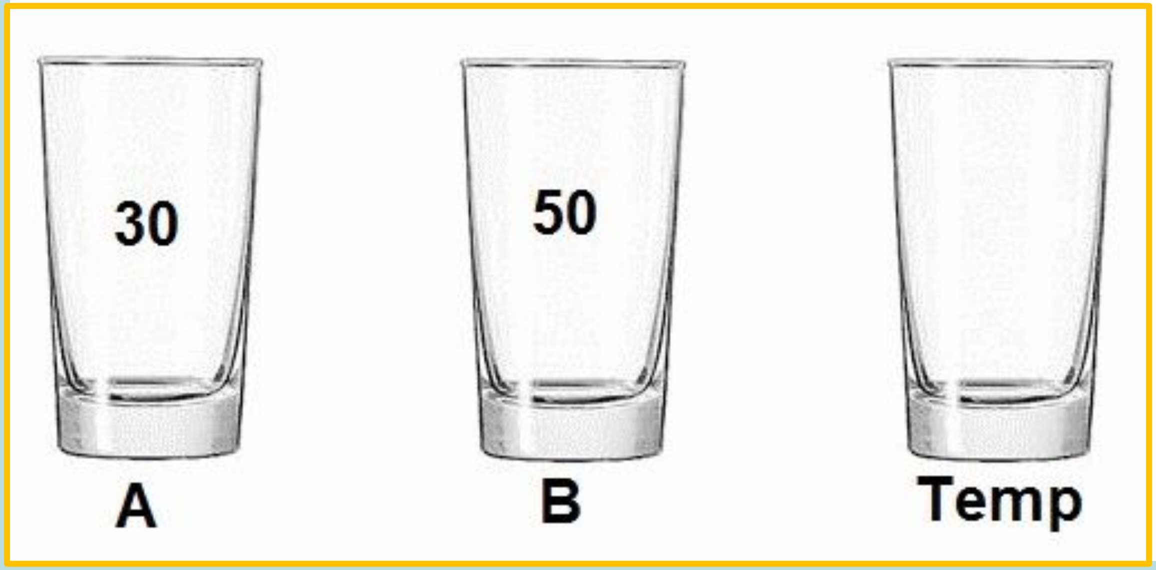


Exercise 2.

Write a program to swap the values of two integer variables using a third variable.

Expected output:

```
Before swapping: a = 50, b = 30
After swapping: a = 30, b = 50
```



Exercise 3.

Fill in the blanks with the correct result:

int a = 10, b = 4, c = 3; float x = 2.5; float y = 1.5;

1. a + b * c = ____________
2. (a + b) * c = ____________
3. a / b = ____________
4. a % b = ____________
5. x + y * c = ____________
6. (x + y) / c = ____________
7. a + (int)x % b = ____________
8. x / y + a - c * b = ____________

Thanks!